

Assignment - 4

Obtain a line that best fit the sample data given in the table. Evaluate the model by finding the standard error.

(x) IDV	Marks scored (Y) DV
1	2
2	4
3	5
4	4
5	5

$$\Rightarrow \text{standard error} = \sqrt{\frac{\sum(\hat{y} - y)^2}{n-2}}$$

$$\bar{x} = \frac{1+2+3+4+5}{5} = 3$$

$$\bar{y} = \frac{2+4+5+4+5}{5} = 4$$

x	y	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$	$\hat{y} = b_0 + b_1 x_i$
1	2	-2	-2	4	4	$2.2 + 0.6 * 1 = 2.8$
2	4	-1	0	1	0	$2.2 + 0.6 * 2 = 3.4$
3	5	0	1	0	0	$2.2 + 0.6 * 3 = 4$
4	4	1	0	1	0	$2.2 + 0.6 * 4 = 4.6$
5	5	2	1	4	2	$2.2 + 0.6 * 5 = 5.2$

$$\hat{y} = b_0 + b_1 x$$

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{6}{10} = 0.6$$

$$b_0 = \bar{y} - b_1 \bar{x} = 4 - 0.6 * 3 = 4 - 1.8 = 2.2$$

$\hat{y} - y_i$	$(\hat{y} - y_i)^2$
0.8	0.64
-0.6	0.36
-1	1
0.6	0.36
0.2	0.04

$$\sum \hat{y} - y_i = \sum (\hat{y} - y_i)^2 = 2.4$$

$$\text{Standard error} = \sqrt{\frac{\sum (\hat{y} - y)^2}{n-2}} = \sqrt{\frac{2.40}{3}} = \sqrt{0.8} \approx 0.894$$

The regression line $\hat{y} = 2.2 + 0.6x$ fits the data with a standard error of about 0.89 marks, meaning predictions from this line typically deviates from the actual score by roughly 0.89 points, which is fairly small relative to range of the data.